

Background and Motivation

- **Low portability:** Some musical instruments are heavy and bulky such as cellos, making them inconvenient to transport from one place to another.
- **Time- and location- constraints:** Playing the cello can be disrupting, particularly during late hours or in a shared space.
- **Limited learning and practicing effectiveness:** There is no feedback provided from musical instruments to quantitatively assess the musical performance

Objectives

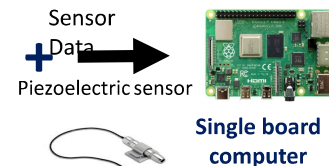
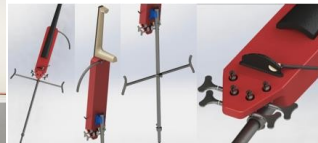
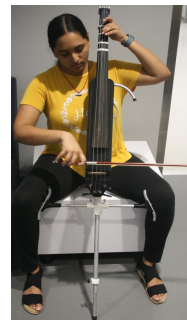
1. Develop an electronic portable cello that preserves the integrity of a regular cello and provides a cello like experience.
2. Develop a quieter cello to allow practicing at night and shared places without jeopardizing the music quality.
3. Create a mobile app with a musical instrument digital interface (MIDI) to access the MMI wirelessly for tuning, synthesizing, and recording of music being played
4. Conduct experimental evaluations on the implemented functions and assess the practicality and believability.

Research Questions

1. Which mechanical design can effectively break down a regular cello to a portable size while keeping the believability of the instrument once assembled and played?
2. Can the cello be made quieter for surrounding but the user can still enjoy the music through a headset?
3. What type of sensors can capture the user inputs (e.g., bowing) with the desired accuracy and resolution?
4. How should the measured electric signals from the sensors be transmitted, processed, and synthesized?
5. How to provide performance feedback to improve the individual's practice session?

Aims and Methods

- **Aim 1:** Compact, quieter, portable mechanical design of a musical instrument
 - **Method:** Identify the most vital and necessary components of the cello; identify the fixed measurements for a 4/4 (Full) size acoustic cello; break down the vital components of the cello to create a compact version that can withstand bending force exerted by the strings on the cello's body [1] and reduce the overall size of the cello to increase ease in portability; eliminate the voice box of the cello and replace the strings with silent strings to reduce loudness.
- **Aim 2:** Sensor selection and instrumentation for digitizing user inputs
 - **Method:** Identify sensors and instrumentation methods to digitize the user inputs in terms of vibration of the strings; Take user inputs, analyze them, and provide feedback on user's playing for learning purposes.
- **Aim 3:** Fabrication, integration of mechanical and electrical components, and programming an on-board computer
 - **Method:** Use a single board computer and algorithms for sensor data acquisition/processing/conversion to MIDI, which then can wirelessly transmit the MIDI data to a mobile device via Bluetooth
- **Aim 4:** Develop a mobile app for user-device interface and control
 - **Method:** Create a user interface (iOS or Android app) to receive the MIDI data from the device, convert it to respective sound, and provide functionalities for real-time playing in headphones and for performance feedback capability



Mechanical design
Sensor selection and implementation

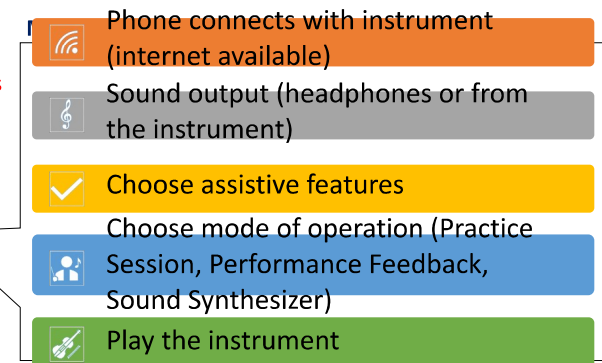
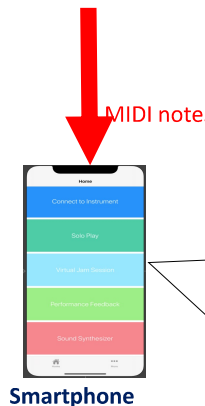
In-ear play



Music

Performance feedback

Improved learning /practicing



POC: Joon-Hyuk Park (joonpark@uct.edu)